

Valuation Project 1: European Static Replication
Due Friday, February 13th
(This is a group assignment. Groups may have up to 3 members.)

The assignment is to:

(a) Use to value a financial instrument from the list below (or of your choice). You should clearly discuss any accuracy concerns you have about your choice of models, discussing your parameters choices and assumptions behind your model .

(b) Select reasonable inputs and estimate the value of the financial instrument, justifying your choices and indicating any concerns that you have about these parameters.

(c) Prepare a short (3-5 pages) written report documenting your implementation of the valuation model and your results. The valuation model and documentation must be completed by the deadline. This should contain:

- A short executive summary of your findings including your estimated value (or range of values)
- A description of the data you collected and used. Be sure to detail the source of your data and any calculations that you had to make.
- A brief description of your static replication (in mathematical notation **NOT** code!).
- A discussion of any sensitivity analysis performed and possible errors in your valuation. Pay attention to volatility choices and dividends here.

Projects will be graded out of 25 points:

5 points for parameter estimation and discussion of data collection.
15 points for the valuation procedure, portfolio construction, precision of valuation etc.
5 points for a discussion of your results, discussion of accuracy of valuation and any necessary sensitivity analysis for uncertain parameters.

Grading will be based on:

- the complexity of the product and/or valuation model.
- how well you carry out the implementation of the valuation model
- your clear description of the model and your analysis.

NOTE: If you wish to choose your own product (see EDGAR notes at the end) you should select a financial instrument that is European in nature and has a least one digital option in the portfolio.

However, there are many financial instruments that are too difficult for you to value at the moment. . I strongly recommend that you identify your financial instrument and come and talk to me about it beforehand, so that I can help you decide whether you have chosen an instrument that is too difficult.

Choice #1: Dual-Directional Buffered PLUS

Russell 2000 Index

https://www.sec.gov/Archives/edgar/data/895421/000183988226000091/ms12688_424b2-00042.htm

Key features:

1. One underlying asset.
2. Digital type payoff

Data: You will need risk-free rates, and dividend yields and implied volatilities for the underlying stock index. You may also want the option vega. These can be obtained from Bloomberg and other sources.

Choice #2: Dual-Directional Buffered PLUS

S&P 500 Index

https://www.sec.gov/Archives/edgar/data/895421/000183988226000097/ms12690_424b2-00048.htm

Key features:

1. One underlying asset.
2. Digital type payoff

Data: You will need risk-free rates, and dividend yields and implied volatilities for the underlying stock index. You may also want the option vega. These can be obtained from Bloomberg and other sources.

Choice #3: Dual-Directional Buffered Securities

Nasdaq 100 Index

https://www.sec.gov/Archives/edgar/data/19617/000121390026000436/ea0271543-01_424b2.htm

Key features:

1. One underlying asset.
2. Digital type payoff

Data: You will need risk-free rates, and dividend yields and implied volatilities for the underlying stock index. You may also want the option vega. These can be obtained from Bloomberg and other sources.

Choice #4: Dual Directional Enhanced Buffered Jump Securities

NVIDIA stock

https://www.sec.gov/Archives/edgar/data/895421/000183988225071813/ms12605_424b2-42210.htm

Key features:

Key features:

1. One underlying asset.
2. Digital type payoff

Data: You will need risk-free rates, and dividend yields (if any!) and implied volatilities for the underlying stock index. You may also want the option vega. These can be obtained from Bloomberg and other sources.

Choice #5: Dual Directional Trigger Jump Securities

Euro Stoxx 50 Index

https://www.sec.gov/Archives/edgar/data/895421/000183988225071715/ms12395_424b2-42119.htm

Key features:

1. One underlying asset.
2. Underlying is NOT denominated in USD and so you will have to be more careful here (and probably talk to me)
3. Digital type payoff

Data: You will need risk-free rates, and dividend yields and implied volatilities for the underlying stock index **and EUR/USD exchange rate volatility** here. You may also want the option vega. These can be obtained from Bloomberg and other sources.